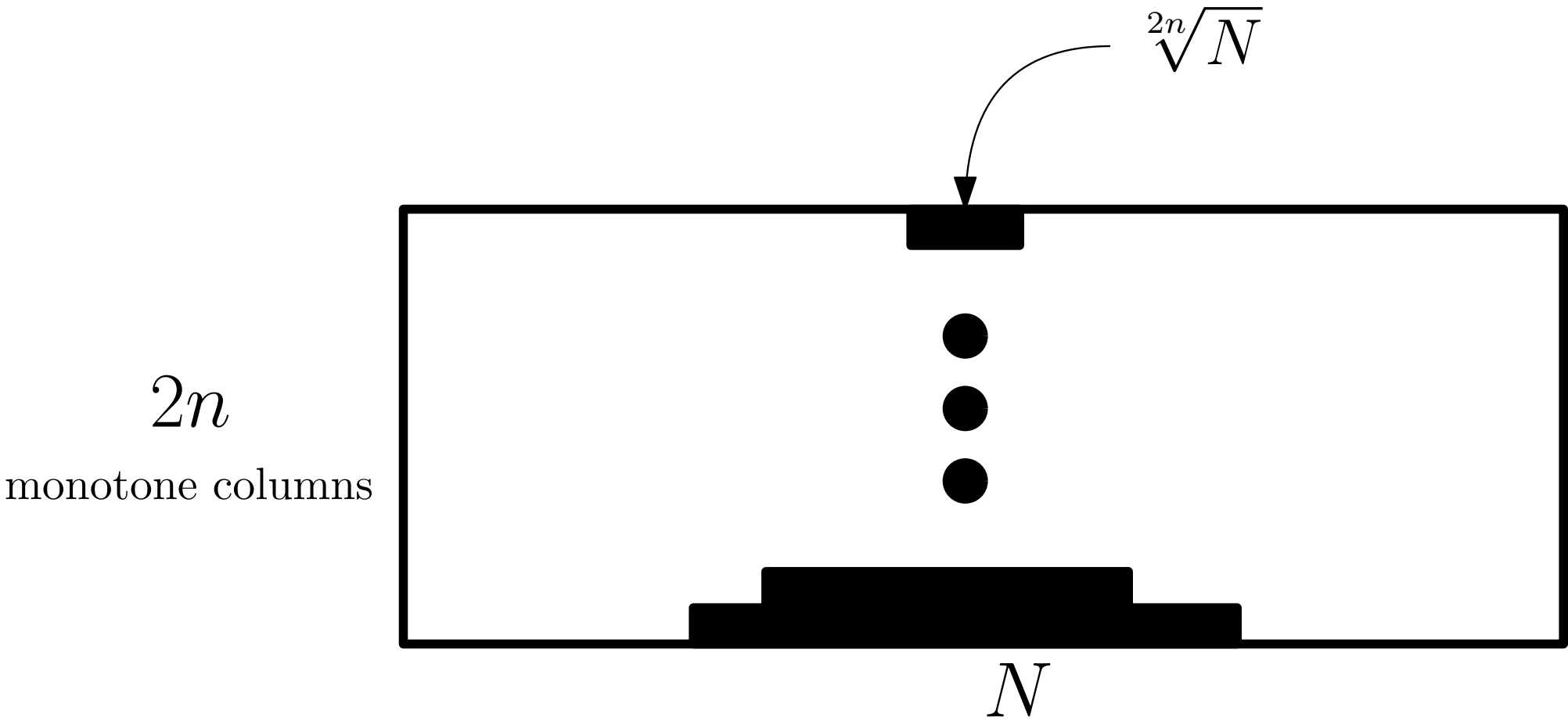

$\sqrt[2n]{N}$
$2n$
monotone columns
$N$